

Methodology: Canadian Residential Electricity Rate Model

Working paper — draft for review Analysis period: 2012–2025 | Geography: 10 Canadian provinces | Updated: April 2026

1. Research question and motivation

Residential electricity rates vary by more than 3× across Canadian provinces at identical consumption levels, and within the same province can swing by 15¢/kWh or more across a three-year window. This analysis asks: what structural forces explain that variation, and what share of observed rate movements can be attributed to each?

The question matters practically. As provinces electrify transportation, space heating, and industry, the level and stability of electricity rates will shape the pace of adoption and the distributional burden of the energy transition. Understanding the mechanisms that generate rate volatility is a precondition for designing policy that reduces it.

The analysis identifies three compounding drivers:

1. **Fuel mix** — the share of consumed electricity generated from fossil fuels, which determines whether a province's cost structure is exposed to commodity price cycles.
 2. **Gas price pass-through** — the extent to which movements in the relevant natural gas benchmark reach retail electricity rates, conditional on fuel mix.
 3. **Market structure** — whether regulatory design transmits commodity shocks immediately (deregulated spot markets) or with a lag and partial absorption (cost-of-service regulation).
-

2. Data sources

2.1 Electricity rates

Source: Hydro-Québec, *Comparison of Electricity Prices in Major North American Cities*, annual editions 2012–2025.

Hydro-Québec compiles this report annually by surveying participating utilities on rates in effect April 1 of each year. The report provides monthly bills and average prices (¢/kWh) at five consumption levels (625, 750, 1,000, 2,000, and 3,000 kWh/month) for Canadian and US cities.

This analysis uses the **average price at 1,000 kWh/month** as the dependent variable. This consumption level is chosen because:

- It approximates median Canadian household monthly electricity consumption.
- It falls in a range where most provincial tiered rate structures are stable.

- It is reported consistently across all years and all provinces in the survey.

Rates are expressed in **cents per kilowatt-hour (¢/kWh)** and exclude taxes and non-utility levies, enabling valid cross-provincial comparison. The 2014 edition provided only monthly bills (CAD\$), not average prices, and is therefore excluded from the main panel, leaving 13 annual observations per province (2012, 2013, 2015–2025).

Cities used and their provincial mapping:

City	Province
Calgary, AB / Edmonton, AB	Alberta (averaged)
Vancouver, BC	British Columbia
Winnipeg, MB	Manitoba
Moncton, NB	New Brunswick
St. John's, NL	Newfoundland and Labrador
Halifax, NS	Nova Scotia
Ottawa, ON / Toronto, ON	Ontario (averaged)
Charlottetown, PE	Prince Edward Island
Montréal, QC	Quebec
Regina, SK	Saskatchewan

Where two cities represent a single province (Alberta, Ontario), the provincial rate is taken as the simple average of the two city rates.

2.2 Generation mix and consumed fossil share

Source (generated mix): Canada Energy Regulator (CER), *Provincial and Territorial Energy Profiles*, accessed 2025; Statistics Canada Table 25-10-0020-01, *Electric power, annual generation by class of producer*.

The generated fossil share for each province-year is constructed as the share of total annual electricity generation attributable to coal, natural gas, and petroleum, expressed as a proportion between 0 and 1.

Critical adjustment — consumed vs. generated mix: The analysis uses consumed fossil share rather than generated fossil share. For provinces that are significant net importers or exporters of electricity, the generated mix misrepresents the fuel content of electricity actually consumed by households. Adjustments are made as follows:

- **British Columbia:** BC Hydro imports approximately 10 TWh per year of off-peak electricity from the US western interconnection, which runs approximately 60% fossil. This raises BC's consumed fossil share from approximately 5–8% (generated) to 12–20% (consumed), depending on year. BC's own hydro exports are excluded from consumption.
- **Prince Edward Island:** PEI generates approximately 99% of its electricity from wind but imports approximately 60–69% of its total consumption from New Brunswick via submarine cable (pre-2017: ~60%; post-2017 interconnection upgrade: ~68–69%). New Brunswick's generation mix runs approximately 27–36% fossil across the period. PEI's consumed fossil share is therefore approximately 18–21%, not the near-zero figure implied by its generation mix.
- **Nova Scotia:** The Maritime Link, commissioned January 2018, delivers approximately 1 TWh per year of Muskrat Falls hydroelectric generation from Newfoundland to Nova Scotia. This marginally reduces NS's consumed fossil share from approximately 70–78% (generated) to approximately 60–66% (consumed) in post-2018 years.
- **Alberta, Manitoba, Quebec, Newfoundland, Saskatchewan, New Brunswick, Ontario:** Net trade flows for these provinces do not materially alter the fossil content of consumed electricity relative to generated electricity, and no adjustment is applied beyond the generation mix data.

Sources for trade adjustments: CER Provincial Energy Profiles; Statistics Canada Annual Electricity Supply and Disposition Survey (Table 25-10-0016-01); PEI Government Energy Strategy 2022; CER *Renewable Energy in Canada* provincial pages.

2.3 Natural gas prices

Two natural gas benchmarks are used, assigned to provinces based on physical supply chain geography.

AECO (Alberta Energy Company / NIT benchmark):

- Relevant for: Alberta, British Columbia, Manitoba, Saskatchewan.
- These provinces are supplied from the Western Canada Sedimentary Basin and price gas against the AECO hub at the Nova Gas Transmission Ltd. (NGTL) system in Alberta.
- **Source:** Alberta Energy and Minerals, *Monthly Natural Gas Reference Price Calculations*, 2012–2025. Monthly values annualised by simple average.
- **Units:** CAD per gigajoule (CAD/GJ).

Henry Hub:

- Relevant for: New Brunswick, Nova Scotia, Newfoundland and Labrador, Prince Edward Island.

- These provinces are supplied via the Maritimes and Northeast Pipeline on contracts linked to the US benchmark Henry Hub in Sabine Pass, Louisiana.
- **Source:** US Energy Information Administration, Henry Hub Natural Gas Spot Price (AHHNGSP annual series), 2012–2025.
- **Units:** USD per million British thermal units (USD/MMBtu), converted to CAD/GJ using the Bank of Canada annual average CAD/USD exchange rate and the unit conversion factor 1 MMBtu = 1.05506 GJ.

Ontario (mixed benchmark):

- Ontario's Dawn Hub in southwestern Ontario closely tracks Henry Hub, but the province also receives AECO-priced gas via the TC Energy Canadian Mainline. The gas price variable for Ontario is set as the simple average of AECO and Henry Hub (both in CAD/GJ) for each year.

Quebec (no exposure):

- Quebec generates over 97% of its electricity from hydroelectric generation with no fuel cost. The gas price variable is set to zero for Quebec, and its rate variation is explained entirely by the province and year fixed effects.

2.4 Market structure classification

Each province is classified into one of three market structure categories based on the institutional arrangements governing wholesale and retail electricity pricing:

Category	Provinces	Description
Deregulated	Alberta	Competitive wholesale pool market; real-time spot prices set by marginal generator; retail rates linked to pool price via regulated default tariffs or direct market contracts.
Hybrid	Ontario, New Brunswick	Partially competitive or regulated-with-market-exposure structures. Ontario has competitive generation with the Global Adjustment mechanism recovering long-term contract costs. NB Power is a regulated monopoly with active market purchases from Quebec and Maine and a fuel adjustment clause.
Regulated	BC, Manitoba, NS, NL, PEI, QC, SK	Vertically integrated regulated monopolies; rates set by provincial utility boards through cost-of-service rate cases; fuel cost shocks absorbed onto utility balance sheet and recovered with a lag.

3. Model specification

3.1 Estimating equation

The main estimating equation is a panel OLS model with province and year fixed effects:

$$\begin{aligned} \text{Rate}_{it} = & \alpha_i + \lambda_t + \beta_1 \cdot \text{Fossil}_{it} + \beta_2 \cdot (\text{Fossil}_{it} \times \text{Dereg}_i) + \beta_3 \cdot (\text{Fossil}_{it} \times \text{Hybrid}_i) \\ & + \beta_4 \cdot \text{Dereg}_i + \beta_5 \cdot \text{Hybrid}_i + \beta_6 \cdot \text{Gas}_{it} + \beta_7 \cdot (\text{Gas}_{it} \times \text{Fossil}_{it}) + \beta_8 \cdot (\text{Gas}_{it} \times \text{Dereg}_i) + \varepsilon_{it} \end{aligned}$$

Where:

- Rate_{it} is the residential average price at 1,000 kWh/month (¢/kWh) in province i , year t .
- α_i is the province fixed effect, absorbing all time-invariant provincial characteristics (geography, hydro endowment, legacy capital stock, regulatory history).
- λ_t is the year fixed effect, absorbing common national shocks (recessions, national policy changes, Canada-wide weather events).
- Fossil_{it} is the consumed fossil share (0–1 scale).
- Dereg_i and Hybrid_i are binary indicators for market structure category (regulated is the omitted reference category).
- Gas_{it} is the annual average gas price in CAD/GJ, assigned by benchmark according to Section 2.3.
- ε_{it} is the idiosyncratic error term.

3.2 Coefficient interpretations

Coefficient	Symbol	Interpretation
Fossil share	β_1	¢/kWh rate change per 1-unit increase in consumed fossil share (0→100%), holding gas price constant, in a regulated province. Identified from within-province changes in fossil share over time.
Fossil × Deregulated	β_2	Additional ¢/kWh per unit fossil share for deregulated provinces, relative to regulated. Captures stronger fossil-to-rate relationship in market-exposed systems.
Fossil × Hybrid	β_3	Same as β_2 for hybrid market provinces.
Deregulated (FE)	β_4	Level rate premium for deregulated provinces, net of fuel mix and gas price effects.
Hybrid (FE)	β_5	Level rate premium for hybrid provinces.

Coefficient	Symbol	Interpretation
Gas price	β_6	¢/kWh rate change per \$1/GJ increase in gas benchmark, at zero fossil share (i.e., for a pure hydro regulated province — expected near zero).
Gas × Fossil share	β_7	Amplification of gas price pass-through by fuel mix. A province with 70% fossil share experiences $\beta_6 + 0.70 \cdot \beta_7$ cents per \$1/GJ gas price movement.
Gas × Deregulated	β_8	Additional gas price pass-through for deregulated provinces above and beyond the fuel mix amplification. Captures the institutional speed and completeness of commodity shock transmission.

3.3 Fixed effects estimation via within-group demeaning

Province and year fixed effects are absorbed using the two-way within-group transformation (Mundlak-style demeaning). For any variable v_{it} :

$$\tilde{v}_{it} = v_{it} - \bar{v}_{i.} - \bar{v}_{.t} + \bar{v}_{..}$$

Where $\bar{v}_{i.}$ is the province mean, $\bar{v}_{.t}$ is the year mean, and $\bar{v}_{..}$ is the grand mean. The transformed variables $\tilde{v}_{it}$ have the within-province, within-year means removed. OLS is then applied to the transformed variables without including province or year dummies explicitly.

This approach is computationally equivalent to including province and year dummy variables in the regression but avoids the degrees-of-freedom cost of explicit dummies and is the standard approach in panel econometrics (Wooldridge 2010, §14.2).

3.4 Standard errors

Standard errors are HC1 heteroskedasticity-robust (White 1980, with the finite-sample correction factor $n/(n - k)$). The residual variance is not assumed constant across provinces with very different rate levels, which makes heteroskedasticity plausible. Province-clustered standard errors would additionally account for serial correlation within provinces and would be the preferred choice in a formal publication with a longer time series; the HC1 choice here reflects the small sample ($n=130$), where cluster robust SEs can be unreliable with only 10 clusters (Cameron and Miller 2015).

4. Driver decomposition

4.1 Decomposition framework

For each province-year observation, the predicted rate is decomposed into four economically labelled components that sum to the total prediction:

$$\text{Rate}_{it} = \underbrace{(\bar{R}_{i\cdot} + \bar{R}_{\cdot t} - \bar{R}_{\cdot\cdot})}_{\text{Baseline}} + \underbrace{\hat{\beta}_1 \tilde{F}_{it} + \hat{\beta}_2 \tilde{F}_{it} D_i + \hat{\beta}_3 \tilde{F}_{it} H_i}_{\text{Fossil mix}} + \underbrace{\hat{\beta}_6 \tilde{G}_{it} + \hat{\beta}_7 \tilde{G}_{it} F_{it}}_{\text{Gas pass-through}} + \underbrace{\hat{\beta}_4 D_i + \hat{\beta}_5 H_i + \hat{\beta}_8 \tilde{G}_{it}}_{\text{Market structure}}$$

Where tildes denote within-transformed variables and bars denote group means.

Baseline: The sum of the province fixed effect and the year fixed effect — what the rate would be if fossil share and gas prices were at their panel-wide means. Captures time-invariant provincial characteristics (hydro endowment, legacy capital, regulatory history) and common year-specific shocks.

Fossil mix contribution: The rate change attributable to within-province variation in consumed fossil share, including the interaction with market structure (because the fossil-to-rate relationship is steeper in deregulated markets). Identifies structural fuel cost exposure independent of what commodity prices are doing in any given year.

Gas price pass-through: The rate change attributable to movements in the relevant gas benchmark, amplified by the province's fossil share. This component is large and volatile for AECO-exposed western provinces in 2019 (AECO collapse) and 2022 (AECO spike), and moderate and smoother for Henry Hub-exposed Atlantic provinces.

Market structure amplification: The portion of rate variation attributable to the institutional speed and completeness of cost shock transmission — the deregulated level premium, the hybrid premium, and the additional gas price pass-through that deregulated markets impose above and beyond what fossil share alone predicts. This component is near zero for all regulated provinces in every year, positive for Ontario and NB in years of gas price elevation, and large and variable for Alberta.

Residual: The difference between the actual rate and the model prediction. Economically interpretable patterns in the residual — systematic underprediction of Alberta in 2022–2023 (scarcity pricing events), overprediction of Ontario in 2020 (rate relief policy), negative-then-positive swing for NS and NB across 2022–2024 (regulatory lag) — are reported and interpreted as confirming evidence for mechanisms the annual model cannot fully capture.

4.2 Identification

The model's causal claims rest on within-province, within-year variation. The province fixed effects absorb all time-invariant confounders — the analysis is not claiming that hydro provinces have lower rates because of their fuel mix (that is a cross-sectional claim the design cannot support); it is claiming that as provinces' fossil shares changed over the 2012–2025 period, their rates moved in ways consistent with the estimated coefficients.

The primary threat to identification is reverse causality: if high rates cause provinces to invest in renewables, reducing fossil share, the fossil share coefficient would be biased toward zero (attenuation), making the results conservative. Simultaneity between gas prices and provincial regulatory decisions is less of a concern because AECO and Henry Hub are national and international benchmarks driven by production and demand factors outside any single province's control.

5. Sample and panel structure

Dimension	Detail
Provinces	10 (all provinces with cities in Hydro-Québec survey)
Years	13 (2012, 2013, 2015–2025; 2014 excluded — average price data unavailable)
Observations	130 province-year cells (balanced panel)
Dependent variable	Residential average price, 1,000 kWh/month (¢/kWh)
Regressors	8 (post within-transformation; province and year FEs absorbed)
Fixed effects	10 province FEs + 13 year FEs = 23 absorbed parameters
Effective DoF	$130 - 8 - 23 = 99$

The panel is balanced — every province has an observation in every year. No imputation is used; the 2014 exclusion removes an entire year rather than specific cells.

6. Limitations and caveats

Sample size. With 10 provinces and 13 years, the panel is small relative to the number of parameters estimated. Interaction terms in particular are identified from limited variation. The directional findings are credible; the magnitudes of the interaction coefficients should be treated as order-of-magnitude estimates rather than precise quantities. Extending the panel to include US state utilities (approximately 40–50 additional cross-sections) would substantially improve precision.

Annual gas prices. The model uses annual average gas prices, which smooth over intra-year volatility. Alberta's pool price dynamics include within-year scarcity pricing events — hours at the \$1,000/MWh cap — that are not captured by an annual AECO average. This causes systematic underprediction of Alberta rates in 2022–2023. Monthly data would improve fit for deregulated provinces.

Regulatory lag. The model uses contemporaneous gas prices rather than lagged ones. For regulated provinces, the mechanism is: gas price rises → utility absorbs cost → files rate case (6–18 months later) → regulator approves increase (6–12 months after filing) → rate increase reaches consumer. A distributed lag specification (gas price at t , $t - 1$, $t - 2$) would improve fit for regulated provinces and produce a cleaner separation between deregulated (immediate pass-through) and regulated (lagged pass-through) behaviour. The current specification's residuals

for NS, SK, and NB show the signature of this lag — negative residuals in the year of the gas price shock, positive residuals one to two years later.

PEI benchmark assignment. PEI's consumed fossil share reflects imports from NB Power, but its electricity rates are set by Maritime Electric's cost recovery of what it pays NB Power — a relationship that depends on contract terms, not just NB's generation costs. The Henry Hub benchmark is a reasonable proxy for Atlantic gas costs but does not perfectly capture the pricing dynamics of NB Power's own fuel contracts.

Ontario's Global Adjustment. Ontario's rate variation from 2014 onwards is substantially driven by the Global Adjustment mechanism, which recovers stranded costs from nuclear and renewable long-term contracts and has no direct relationship to gas prices or consumed fossil share. The model systematically misfits Ontario as a result. Ontario arguably requires its own structural model. Coding it as a separate indicator rather than "Hybrid" improves fit slightly but does not resolve the fundamental issue.

Capital costs and debt. The province fixed effects absorb long-run differences in capital cost structures (Quebec's paid-off dams, Newfoundland's Muskrat Falls debt burden), but within-province changes in capital costs are not modelled. The sharp increase in Newfoundland rates starting in 2021 — driven by Muskrat Falls debt service — is partially picked up by year effects and partially by the residual, but is not attributed to a named driver.

No demand-side controls. The model has no controls for heating degree days, provincial GDP, or population. Demand-side shocks that affect both electricity consumption and rate recovery are a source of omitted variable bias. Cold winters increase both demand and the fossil share of actual dispatch, creating a potential spurious positive correlation between fossil consumption and rates within a year.

7. Data availability and replication

All data used in this analysis are publicly available from the sources cited. The Python replication script (`electricity_model.py`) produces the full Excel output from raw data and requires no proprietary data sources.

Dataset	Source	URL
Electricity rates	Hydro-Québec	hydroquebec.com/documents-data/official-publications
Generation mix	CER / StatCan	cer-rec.gc.ca; statcan.gc.ca (Table 25-10-0020-01)
Electricity trade	StatCan AELE	statcan.gc.ca (Table 25-10-0016-01)
AECO prices	Alberta Energy & Minerals	alberta.ca/alberta-natural-gas-reference-price
Henry Hub prices	EIA	eia.gov (AHHNGSP annual series)
CAD/USD exchange rate	Bank of Canada	bankofcanada.ca

8. References

Cameron, A.C. and Miller, D.L. (2015). "A practitioner's guide to cluster-robust inference." *Journal of Human Resources*, 50(2), 317–372.

Canada Energy Regulator (2025). *Provincial and Territorial Energy Profiles*. Ottawa: CER.

Hydro-Québec (2012–2025). *Comparison of Electricity Prices in Major North American Cities*. Montréal: Hydro-Québec.

Statistics Canada (2025). Table 25-10-0020-01, *Electric power, annual generation by class of producer*. Ottawa: StatCan.

Statistics Canada (2025). Table 25-10-0016-01, *Electric power generation, monthly receipts, deliveries and availability*. Ottawa: StatCan.

US Energy Information Administration (2026). *Henry Hub Natural Gas Spot Price (AHHNGSP)*. Washington: EIA.

White, H. (1980). "A heteroskedasticity-consistent covariance matrix estimator and a direct test for heteroskedasticity." *Econometrica*, 48(4), 817–838.

Wooldridge, J.M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. Cambridge: MIT Press.

This document should be read alongside the Python replication script and Excel output file. All coefficient estimates, fit statistics, and decomposition figures reported here are produced directly

by the script from the raw data and can be reproduced by running `python
electricity_model.py`.